

A Compact Narrative for Prediction-Relevant Operator Mechanisms

Fernando Ruiz Mazo

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1 The Question

The starting point is a simple ambiguity. A transformer trained for in-context linear regression may make predictions close to linear-kernel ridge regression, but pointwise accuracy on one sampled label vector does not identify the computation being used. Many algorithms can agree with ridge regression on the realized labels while responding differently to nearby counterfactual labels.

The note therefore changes the object of study. Instead of asking only whether the transformer predicts the right targets for one sampled context, it freezes the input geometry and perturbs only the context labels. The first question becomes:

When labels are changed infinitesimally, does the transformer respond like the ridge operator?

This operator-response test is only the entry point. The paper then asks a stronger mechanistic question: if the transformer has a ridge-like local response, when does a hidden layer contain a compact context-side subspace sufficient to reconstruct the ridge operator at the prediction-risk scale of the task? Causal use is then a separate intervention question: after a sufficient subspace has been identified, do interventions preserving or removing that subspace preserve or damage the transformer’s response?

The resulting claim is stronger than behavioral imitation but narrower than global equality. In the successful regimes, the transformer locally realizes the task-distributed ridge label-response operator, and hidden response spans contain low-dimensional subspaces whose induced reduced KRR operators recover exact KRR up to a predeclared prediction-risk tolerance. The number of required directions is computed before looking at activations. The locality matters: the tests concern the sampled geometry, the chosen perturbation scale, and especially task-label directions rather than all possible label perturbations.

2 Fixed Geometry Makes Ridge Regression an Operator

Fix one in-context episode with context and target inputs

$$X_{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times d_x}, \quad X_{\text{tgt}} \in \mathbb{R}^{n_{\text{tgt}} \times d_x}.$$

The fixed geometry is

$$G = (X_{\text{ctx}}, X_{\text{tgt}}).$$

During the operator tests, G is held fixed. The context inputs do not move, the target inputs do not move, and the only varying object is the context-label vector

$$y = (y_1, \dots, y_{n_{\text{ctx}}}) \in \mathbb{R}^{n_{\text{ctx}}}.$$

Define

$$K = X_{\text{ctx}} X_{\text{ctx}}^\top, \quad K_t = X_{\text{tgt}} X_{\text{ctx}}^\top, \quad A = K + \sigma^2 I.$$

For linear-kernel ridge regression, the dual coefficients and target predictions are

$$\alpha^\star = A^{-1}y, \quad y^\star = K_t \alpha^\star = K_t A^{-1}y.$$

Thus, once G is fixed, ridge regression is a linear map from context labels to target predictions:

$$T_G := K_t A^{-1}, \quad T_G : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}.$$

The transformer defines a map on the same spaces:

$$F_\theta^G : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}, \quad F_\theta^G(y) = \hat{y}_\theta.$$

When the geometry is clear, write T and F . The central comparison is not merely $F(y) \approx Ty$ for one y . It is whether the local response of F to label perturbations matches the operator T .

3 Context-Label Directions, Local Operators, and Additivity

A context-label direction is a vector

$$u \in \mathbb{R}^{n_{\text{ctx}}}.$$

It assigns one scalar to each context example. Perturbing in direction u means changing labels as

$$y + \varepsilon u = (y_1 + \varepsilon u_1, \dots, y_{n_{\text{ctx}}} + \varepsilon u_{n_{\text{ctx}}}).$$

This is a direction over context positions. It is not a direction in input-feature space and not a direction in target space. If there are 47 context examples, these directions live in $\mathbb{R}^{47}$.

Pointwise agreement checks only the value of two maps at one label vector:

$$F(y) \approx Ty.$$

Since fixed-geometry ridge regression is linear, its Jacobian is the operator itself:

$$J_T(y) = T \quad \text{for every } y.$$

The transformer map F can be nonlinear in the labels. The weakest local version of “the same label-to-prediction rule” is therefore equality of first-order responses:

$$J_F(y)u \approx Tu.$$

Equivalently, for small label perturbations,

$$F(y + \delta) \approx F(y) + T\delta.$$

The experiments estimate the Jacobian action with the central finite difference

$$D_\varepsilon F(y)[u] := \frac{F(y + \varepsilon u) - F(y - \varepsilon u)}{2\varepsilon}.$$

The symmetric form is used because it cancels the leading first-order bias of a one-sided difference: for a smooth map, the error is $O(\varepsilon^2)$ rather than $O(\varepsilon)$. It also treats positive and negative label

perturbations in the same direction symmetrically, which is important when the goal is to estimate the local tangent response rather than a one-sided effect. The behavioral operator test asks whether

$$D_\varepsilon F(y)[u] \approx Tu$$

on held-out label directions.

This is still a local statement. Matching the value and first derivative at one point does not imply global equality. For example, $f(x) = x + x^2$ and $g(x) = x$ have the same value and derivative at $x = 0$, but differ away from zero. Thus derivative matching supports tangent-level agreement, not $F(y') = Ty'$ for all possible labels.

The probe distribution specifies which counterfactual label directions are tested. The main probes are task-label probes:

$$u \sim \mathcal{N}(0, A).$$

This is the natural law because, under the matched linear task

$$y = X_{\text{ctx}}\beta + \eta, \quad \beta \sim \mathcal{N}(0, I), \quad \eta \sim \mathcal{N}(0, \sigma^2 I),$$

conditioning on the fixed geometry leaves only β and η random, so

$$y \mid G \sim \mathcal{N}(0, X_{\text{ctx}}X_{\text{ctx}}^\top + \sigma^2 I) = \mathcal{N}(0, A).$$

Equivalently, the conditional label covariance is

$$\text{Cov}(y \mid G) = A.$$

Thus $u \sim \mathcal{N}(0, A)$ samples counterfactual label perturbations with the same covariance geometry as labels generated by the task, after the input geometry has been fixed. In this precise sense, task probes are the directions the task distribution itself emphasizes. In whitened coordinates, they are $u = A^{1/2}z$, $z \sim \mathcal{N}(0, I)$. The experiments also use isotropic probes

$$u \sim \mathcal{N}(0, I),$$

which test arbitrary context-label directions equally. The distinction is central: the claim is task-local because the transformer is close to ridge on task-label probes but not on arbitrary isotropic probes. Build probes are used to construct subspaces or fitted controls; held-out probes are used for reported errors.

For a held-out probe set $U = \{u_1, \dots, u_m\}$ and a fixed linear operator $S : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}$, define

$$E_\varepsilon(F, S; U) := \left(\frac{\sum_{j=1}^m \|D_\varepsilon F(y)[u_j] - Su_j\|_2^2}{\sum_{j=1}^m \|Tu_j\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

When $S = T$, this measures whether the transformer’s finite-difference response matches exact ridge regression. The denominator normalizes by the ridge-response energy on the same probes, so an error of 0.01 means a root-mean-square discrepancy of about one percent of the typical ridge response size.

Proposition 1 (Transformer errors are not bounded by 1). *For a finite probe set U , write*

$$R_U^2 = \sum_j \|Tu_j\|_2^2.$$

If $D_\varepsilon F(y)[u_j] = 0$ for all probes, then

$$E_\varepsilon(F, T; U) = \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}} \leq 1.$$

If $D_\varepsilon F(y)[u_j] = -Tu_j$, then the same calculation gives a value at most 2. More generally, if $D_\varepsilon F(y)[u_j] = cTu_j$, then

$$E_\varepsilon(F, T; U) = |c - 1| \frac{R_U}{(R_U^2 + \eta_{\text{num}})^{1/2}},$$

so transformer-response errors have no universal upper bound.

Proof. Substitute the three proposed finite-difference responses into the definition of $E_\varepsilon(F, T; U)$. The last expression diverges as $|c| \rightarrow \infty$ whenever the ridge response energy R_U^2 is nonzero. $\square$

Because ridge is linear, its higher-order derivatives are zero. We do not try to estimate all higher-order derivatives of the transformer. Instead, we check whether nonlinear interactions are small at the tested perturbation scale. Ridge has

$$T(u + v) = Tu + Tv.$$

A nonlinear transformer need not satisfy

$$D_\varepsilon F(y)[u + v] \approx D_\varepsilon F(y)[u] + D_\varepsilon F(y)[v].$$

If this approximate additivity failed badly, then $E_\varepsilon(F, T)$ would mix two effects: failure to be ridge and failure to have a stable local linear response.

For paired directions $V = \{(u_j, v_j)\}_{j=1}^m$, define the second mixed finite difference

$$C_j = F(y + \varepsilon u_j + \varepsilon v_j) - F(y + \varepsilon u_j - \varepsilon v_j) - F(y - \varepsilon u_j + \varepsilon v_j) + F(y - \varepsilon u_j - \varepsilon v_j),$$

and the additivity defect

$$A_\varepsilon(F; V) := \left(\frac{\sum_{j=1}^m \|C_j\|_2^2}{\sum_{j=1}^m \|T(\varepsilon u_j + \varepsilon v_j)\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

For any affine function of the labels, $C_j = 0$ exactly. Because the defect is normalized by ridge-response energy, values below one mean the non-additive interaction is smaller than the typical ridge response on the tested paired probes; values around one mean it is ridge-scale; values above one mean the interaction is larger than the ridge response, so a local-linear interpretation is unreliable. This diagnostic does not identify the operator. It only says whether a local operator comparison is meaningful at the chosen ε and probe law.

4 Activation Subspaces and Prediction-Risk Rank

The previous section gives behavioral certificates:

$$F(y) \approx Ty, \quad D_\varepsilon F(y)[u] \approx Tu.$$

These are necessary, but they are still input-output facts. They do not say whether the transformer represents the directions needed for ridge regression, or whether those directions are used. The

mechanistic question is: does the residual stream expose a compact set of context-position directions sufficient for the prediction-relevant ridge computation?

This is the right internal object to look for because ridge uses context-position dual variables. For fixed geometry,

$$\alpha^*(y) = A^{-1}y \in \mathbb{R}^{n_{\text{ctx}}}, \quad Ty = K_t \alpha^*(y).$$

The dual coordinate α_i is indexed by the i -th context example. The context residual stream has the same context axis. At layer ℓ ,

$$H_\ell^{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times D},$$

where D is the residual width. Rows are context tokens, and the k -th column $H_\ell^{\text{ctx}}[:, k]$ is a scalar pattern over context positions, hence a vector in $\mathbb{R}^{n_{\text{ctx}}}$. Linear combinations of residual-stream columns therefore define candidate directions in the same space as ridge dual variables.

Let

$$Q = [q_1, \dots, q_r] \in \mathbb{R}^{n_{\text{ctx}} \times r}$$

be a basis matrix for such candidate context-position directions. The subspace is the column span

$$\text{span}(Q) = \{Qc : c \in \mathbb{R}^r\} \subseteq \mathbb{R}^{n_{\text{ctx}}},$$

so Q is both a matrix representation and, by abuse of language, the subspace it spans. We use an A -orthonormal basis,

$$Q^\top A Q = I,$$

because the quadratic part of the ridge dual objective is $\frac{1}{2} \alpha^\top A \alpha$. Thus the natural size of a dual displacement is $\alpha^\top A \alpha$, and two directions are orthogonal in the ridge geometry when their cross-term in this quadratic energy vanishes.

Now define what it means for $\text{span}(Q)$ to be useful for ridge. This is a restriction of the dual solution, not an ordinary restriction of the input label vector. Full ridge solves

$$\alpha^*(y) = \arg \min_{\alpha \in \mathbb{R}^{n_{\text{ctx}}}} \left\{ \frac{1}{2} \alpha^\top A \alpha - \alpha^\top y \right\},$$

whose first-order condition is $A\alpha - y = 0$, hence $\alpha^* = A^{-1}y$. To restrict the dual solution to $\text{span}(Q)$, write $\alpha = Qc$. Plugging this into the same dual objective gives

$$\frac{1}{2} c^\top Q^\top A Q c - c^\top Q^\top y.$$

Since $Q^\top A Q = I$, this is

$$\frac{1}{2} c^\top c - c^\top Q^\top y.$$

Differentiating with respect to c gives $c - Q^\top y = 0$, so

$$c^* = Q^\top y.$$

Therefore the restricted dual solution is

$$\alpha_Q(y) = Q Q^\top y.$$

Without the A -orthonormal convention, the same calculation would give

$$\alpha_Q(y) = Q(Q^\top A Q)^{-1} Q^\top y.$$

Predictions are still read out through the context-target kernel matrix K_t , so the reduced ridge operator induced by Q is

$$T_Q := K_t Q Q^\top.$$

This is the bridge from residual-stream directions to an operator-level certificate. We are not fitting an arbitrary probe. We are asking whether the directions exposed by the residual stream are enough so that ridge, restricted to those directions, still predicts like full ridge.

The reason this sufficiency test is prediction-relevance weighted rather than global is that not every label direction matters equally for prediction. Conditional on the fixed geometry,

$$y \mid G \sim \mathcal{N}(0, A).$$

So the task naturally emphasizes perturbations

$$u \sim \mathcal{N}(0, A),$$

not arbitrary Euclidean directions. A direction is task-relevant when it is likely under this covariance geometry and when ridge produces a non-negligible target response along it. This leads to a purely ridge-theoretic benchmark: before looking inside the transformer, ask how many directions are needed by the best possible reduced ridge response on task-distributed labels.

To turn this into a prediction-scale benchmark, we need a reference risk. The target quantity is not the noisy context-label vector y . It is the vector of latent, noise-free function values at the target inputs. Write

$$f_{\text{tgt}} = (f(x_1^{\text{tgt}}), \dots, f(x_{n_{\text{tgt}}}^{\text{tgt}})) \in \mathbb{R}^{n_{\text{tgt}}},$$

and let K_{tt} be the target-target kernel matrix. In the linear case, $K_{tt} = X_{\text{tgt}} X_{\text{tgt}}^\top$. Under the matched GP/KRR model,

$$\begin{bmatrix} y \\ f_{\text{tgt}} \end{bmatrix} \mid G \sim \mathcal{N}\left(0, \begin{bmatrix} A & K_t^\top \\ K_t & K_{tt} \end{bmatrix}\right),$$

because $y = f_{\text{ctx}} + \eta$ includes label noise while f_{tgt} is the noise-free target function. Conditioning this Gaussian gives

$$\mathbb{E}[f_{\text{tgt}} \mid y, G] = K_t A^{-1} y = T y$$

and

$$\text{Cov}(f_{\text{tgt}} \mid y, G) = K_{tt} - K_t A^{-1} K_t^\top.$$

Thus exact KRR is the posterior mean predictor of the latent target values. It still has nonzero risk, because the noisy context labels do not determine the target function values exactly. Define this irreducible KRR posterior prediction risk by

$$R_{\text{KRR}}(G) := \mathbb{E} \|f_{\text{tgt}} - T y\|_2^2 = \text{tr}\left(K_{tt} - K_t A^{-1} K_t^\top\right),$$

where the expectation is conditional on the fixed geometry G .

Now compare exact KRR with any linear approximation S to T . Decompose

$$f_{\text{tgt}} - S y = (f_{\text{tgt}} - T y) + (T - S) y.$$

The first term is the posterior residual after exact KRR, and the second term is a function of the observed labels y . Since $T y = \mathbb{E}[f_{\text{tgt}} \mid y, G]$, the posterior residual is orthogonal to every function of y . Therefore the cross-term vanishes and

$$\mathbb{E} \|f_{\text{tgt}} - S y\|_2^2 = R_{\text{KRR}}(G) + \mathbb{E}_{y \sim \mathcal{N}(0, A)} \|(T - S) y\|_2^2.$$

The second term is the extra prediction risk caused by using S instead of exact KRR. Thus the natural scale for deciding whether a reduced operator is good enough is not zero operator error, but excess error relative to the irreducible risk already incurred by exact KRR. Define

$$\rho_G(S) := \frac{\mathbb{E}_{y \sim \mathcal{N}(0, A)} \|(T - S)y\|_2^2}{R_{\text{KRR}}(G) + \eta_{\text{num}}}.$$

This asks how much prediction risk is added by using S instead of exact KRR.

The next question is how many source directions are needed in principle. This should be answered before looking at the transformer. We therefore ask a purely KRR question: among all rank- r linear responses to task-distributed labels, how much extra prediction risk is unavoidable if we keep only r directions?

Whiten the task-label perturbations:

$$u = A^{1/2}z, \quad z \sim \mathcal{N}(0, I).$$

Here z is an isotropic coordinate system for labels drawn from the task covariance A . In these coordinates, the exact KRR response is

$$Tu = K_t A^{-1} A^{1/2} z = K_t A^{-1/2} z.$$

Define

$$C_G := K_t A^{-1/2}.$$

Thus C_G is not a new predictor. It is the same KRR operator written in coordinates where the task-label distribution is spherical. This distinction matters. Singular values of T measure response size to raw Euclidean label directions $u \sim \mathcal{N}(0, I)$, which is not the task law. Singular values of C_G measure response size to label directions that the task actually generates, $u \sim \mathcal{N}(0, A)$.

Let

$$C_G = U \text{diag}(s_i) V^\top, \quad s_1 \geq s_2 \geq \dots \geq 0.$$

If z points along the right singular vector v_i , the KRR prediction moves in the corresponding target direction with size s_i . Thus s_i^2 is the squared prediction-response energy carried by the i -th task-label mode. Since z is isotropic,

$$\mathbb{E} \|C_G z\|_2^2 = \sum_i s_i^2.$$

Now suppose we approximate exact KRR by a rank- r linear operator. In whitened coordinates this means approximating C_G by a rank- r matrix. By Eckart–Young, the best possible rank- r approximation keeps the r largest singular directions. The unavoidable task-law mean squared operator error is the discarded tail

$$\sum_{i>r} s_i^2.$$

Equivalently, the best possible rank- r predictor has total prediction risk

$$R_{\text{KRR}}(G) + \sum_{i>r} s_i^2.$$

This is why the tail is compared to $R_{\text{KRR}}(G)$, not to zero and not only to the total response energy $\sum_i s_i^2$. The practical question is whether throwing away the tail changes prediction performance by an amount that is negligible relative to the uncertainty already left by exact KRR.

Definition 1 (Prediction-risk rank). *For fixed geometry G and tolerance $\alpha > 0$, define*

$$r_\alpha(G) := \min \left\{ r : \sum_{i>r} s_i^2 \leq \alpha R_{\text{KRR}}(G) \right\}.$$

In the experiments below we write $r_T := r_{5\%}$, i.e. $\alpha = 0.05$. This means that r_T is the smallest rank for which an optimally chosen reduced KRR response would have prediction risk at most

$$(1.05) R_{\text{KRR}}(G).$$

Thus r_T is not the algebraic rank of T , and it is not the number of nonzero singular values. It is the number of task-relevant KRR directions needed to be prediction-equivalent to exact KRR at the declared tolerance. Directions beyond r_T may be mathematically nonzero, but their combined effect on target prediction is below the chosen risk budget.

This rank is an intrinsic finite-episode benchmark. If $\dim \text{span}(Q) < r_T$, then no rank- $\dim \text{span}(Q)$ operator, even an optimally aligned one, can reach the five-percent prediction-risk tolerance. If $\dim \text{span}(Q) \geq r_T$, success is still not automatic: the span must align with the large right singular directions of C_G , because those are the task-label modes that exact KRR uses for prediction.

For the linear kernel,

$$\text{rank}(C_G) = \text{rank}(T) = \text{rank}(K_t) \leq \min\{d_x, n_{\text{ctx}}, n_{\text{tgt}}\},$$

because A is invertible and $K_t = X_{\text{tgt}} X_{\text{ctx}}^\top$. Hence

$$r_T(G) \leq \min\{d_x, n_{\text{ctx}}, n_{\text{tgt}}\}.$$

In the flat random linear geometries used in the small- d_x sweep, this upper bound is typically attained at the five-percent risk tolerance. Writing $X_{\text{ctx}} = P \Sigma V^\top$,

$$C_G = X_{\text{tgt}} V \text{diag} \left(\frac{\sigma_i}{\sqrt{\sigma_i^2 + \sigma^2}} \right) P^\top.$$

With $n_{\text{ctx}} = 47$ and $\sigma^2 = 0.1$, the context singular values are typically much larger than the ridge noise, so the shrinkage factors are close to one. When $d_x \leq n_{\text{tgt}} = 16$, the target-side factor $X_{\text{tgt}} V$ is generically full column rank, so the prediction-risk rank is empirically close to d_x . In the $d_x = 30$ stress case, $n_{\text{tgt}} = 16$ caps the prediction-risk rank.

5 Nonlinear Kernels and the Sample-Size Elbow

The prediction-risk rank definition does not rely on the linear kernel. For a general kernel, replace

$$K = X_{\text{ctx}} X_{\text{ctx}}^\top, \quad K_t = X_{\text{tgt}} X_{\text{ctx}}^\top$$

by $K_{ij} = k(x_i, x_j)$ and $(K_t)_{aj} = k(x_a^{\text{tgt}}, x_j)$, keep $A = K + \sigma^2 I$, and again form

$$C_G = K_t A^{-1/2}.$$

The singular spectrum of C_G is the finite-episode ridge response spectrum on task-distributed label perturbations. Its tail, compared with R_{KRR} , defines $r_T(G)$. It is the number of directions a

mechanism must represent, and align with, to implement KRR at the prediction-risk tolerance on that episode.

For nonlinear kernels, r_T should not be read as an input-dimension count. The finite matrix still obeys

$$\text{rank}(C_G) \leq \min\{n_{\text{ctx}}, n_{\text{tgt}}\},$$

but the population source of the rank is the Mercer spectrum of the kernel under the input law ρ :

$$k(x, x') = \sum_{j \geq 0} \mu_j \phi_j(x) \phi_j(x'), \quad \mu_0 \geq \mu_1 \geq \dots \geq 0.$$

With $n = n_{\text{ctx}}$, ridge resolves mode j with visible energy approximately

$$e_j(n) \asymp \mu_j \frac{n\mu_j}{n\mu_j + \sigma^2} = \frac{\mu_j^2}{\mu_j + \sigma^2/n}.$$

The multiplier $n\mu_j/(n\mu_j + \sigma^2)$ is a statistical visibility factor. Modes satisfying

$$n\mu_j \ll \sigma^2$$

are below the ridge noise floor and contribute only weakly to the label-response operator. Modes satisfying $n\mu_j \gtrsim \sigma^2$ are visible. Thus the population analogue of the prediction-risk rank is

$$r_*(n; \alpha) := \min \left\{ r : \sum_{j > r} e_j(n) \leq \alpha R_{\text{KRR}}^{\text{pop}}(n) \right\}.$$

Here $R_{\text{KRR}}^{\text{pop}}(n)$ is the population posterior prediction risk at context size n .

This gives the right reading of finite ranks:

$$r_T(G) \approx \min\{n_{\text{ctx}}, n_{\text{tgt}}, r_*(n_{\text{ctx}}; 0.05)\},$$

up to finite-sample fluctuations. The first two terms are sequence-design caps. They are not intrinsic limitations of the kernel; they can be raised by using more context or target points. The third term is intrinsic to the chosen kernel, input distribution, noise level, context size, and tolerance. It is the prediction-relevant rank of the population-visible KRR operator.

This is the sample-size elbow. Increasing n_{ctx} lowers the visibility threshold σ^2/n_{ctx} and can reveal additional Mercer modes. Increasing n_{tgt} does not reveal new population modes, but it prevents the finite diagnostic from being target-capped. To train or certify a transformer as implementing the full visible KRR operator in a regime, one should choose sequence lengths so that

$$\min\{n_{\text{ctx}}, n_{\text{tgt}}, r_*(n_{\text{ctx}}; 0.05)\} = r_*(n_{\text{ctx}}; 0.05).$$

Otherwise the model may succeed on, and the diagnostics may certify only, a capped low-rank slice of KRR rather than the full prediction-relevant operator. Since $r_*(n; 0.05)$ depends only on the kernel spectrum, input law, noise level, context size, and tolerance, it can be estimated before training and used to choose the context and target lengths.

Proposition 2 (Prediction-risk excess for T_Q). *Let $u = A^{1/2}z$, $z \sim \mathcal{N}(0, I)$, and let $Q^\top A Q = I$. With*

$$C = K_t A^{-1/2}, \quad W = A^{1/2} Q, \quad P_Q = W W^\top,$$

the reduced-operator excess risk satisfies

$$\rho_G(T_Q) = \frac{\|C(I - P_Q)\|_F^2}{R_{\text{KRR}}(G) + \eta_{\text{num}}}.$$

If W spans the top r right singular directions of C , then

$$\rho_G(T_Q) = \frac{\sum_{i>r} s_i^2}{R_{\text{KRR}}(G) + \eta_{\text{num}}}.$$

Proof. Since $W = A^{1/2}Q$, $W^\top W = Q^\top A Q = I$. For $u = A^{1/2}z$,

$$Tu = K_t A^{-1} A^{1/2} z = Cz, \quad T_Q u = K_t Q Q^\top A^{1/2} z = K_t A^{-1/2} W W^\top z = C P_Q z.$$

Taking expectation over z gives

$$\mathbb{E}\|(T - T_Q)u\|_2^2 = \|C(I - P_Q)\|_F^2.$$

The singular-value expression follows from Eckart–Young: among all rank- r right projections, the top right singular subspace of C minimizes the Frobenius residual, leaving $\sum_{i>r} s_i^2$. $\square$

Remark 1 (Reading the numbers). *The number $\rho_G(S)$ is an excess-risk ratio. Thus $\rho_G(S) = 0.05$ means that using S instead of exact KRR adds five percent of the KRR posterior prediction risk. Values near zero mean the reduced operator is prediction-equivalent to KRR at the chosen tolerance. Values near or above one mean that the reduction adds error comparable to, or larger than, the irreducible KRR risk. Transformer response errors $E_\varepsilon(F, T)$ and $E_\varepsilon(F, T_Q)$ are different finite-difference diagnostics and are not bounded by this risk scale.*

The mechanism argument therefore needs three distinct checks:

$$E_\varepsilon(F, T; U) \text{ small}, \quad \rho_G(T_Q) \text{ small}, \quad E_\varepsilon(F, T_Q; U) \text{ small}.$$

The first is behavioral. The second says the residual-stream subspace supports ridge in the prediction-risk sense. The third says the transformer’s local response follows the reduced operator induced by that same subspace. The decomposition

$$D_\varepsilon F(y)[u] - Tu = (D_\varepsilon F(y)[u] - T_Q u) + (T_Q u - Tu)$$

keeps these claims separate.

It remains to describe how the residual-stream subspace is extracted. The experiments use two final-state constructions in this section, and the next section introduces the causal layerwise construction. Given any context-pattern matrix $M \in \mathbb{R}^{n_{\text{ctx}} \times p}$, compute the A -weighted SVD

$$A^{1/2}M = U\Sigma V^\top,$$

keep the selected left singular directions, and map back:

$$Q(M) := A^{-1/2}U_{\text{kept}}.$$

Then $Q(M)^\top A Q(M) = I$. The kept singular directions are those above the numerical threshold or within the predeclared rank budget. The prediction-risk rank tells how many ridge-relevant directions should be enough in principle; the experiments then ask whether the residual-stream directions achieve small $\rho_G(T_Q)$ around that scale.

The raw final-state extractor uses the unperturbed final residual stream,

$$M_{\text{raw}} = H_L^{\text{ctx}}, \quad Q_{\text{raw}} := Q(M_{\text{raw}}).$$

It asks an availability question: which context-position directions are linearly visible in the final residual stream, and are enough of them present to support the prediction-relevant ridge operator?

The response-only extractor uses finite-difference changes in the final residual stream,

$$Z_L(u) = \frac{H_L^{\text{ctx}}(y + \varepsilon u) - H_L^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}, \quad M_{\text{resp}} = [Z_L(u_1), \dots, Z_L(u_m)],$$

and sets

$$Q_{\text{resp}} := Q(M_{\text{resp}}).$$

It asks the stricter question: among the directions that actually move when labels are perturbed, is there a context-position subspace whose induced T_Q has small prediction-risk excess?

In both constructions Q is frozen after extraction. If the basis were re-extracted for each held-out perturbation, it would not define one stable reduced operator. The certificate is that a fixed subspace extracted from residual-stream activations induces a fixed T_Q that agrees with ridge and with the transformer’s held-out local response.

6 Layerwise Availability Construction

The final-state certificate can show that a useful subspace is present at the end of the computation, but it does not explain how that subspace appears across layers. The revised layerwise construction is a post-training diagnostic for this question. It asks, at each layer, whether the layer’s label-response span contains a subspace of approximately r_T dimensions whose induced reduced KRR operator reaches the prediction-risk tolerance.

This replaces the older primary emphasis on one-at-a-time causal direction selection. Individual ablation is a useful causal-use diagnostic, but it is not the cleanest definition of emergence: a distributed representation can contain the right subspace while no single arbitrary basis direction is individually necessary. The layerwise sufficiency construction first finds a KRR-sufficient subspace; causal intervention should then test the selected subspace as a whole.

At layer ℓ , define the hidden-state finite-difference response

$$Z_\ell(u) := \frac{H_\ell^{\text{ctx}}(y + \varepsilon u) - H_\ell^{\text{ctx}}(y - \varepsilon u)}{2\varepsilon}.$$

Stacking these matrices over build probes gives

$$M_\ell = [Z_\ell(u_1), \dots, Z_\ell(u_m)].$$

The columns of M_ℓ are context-position patterns that are visible at layer ℓ when context labels move. An A -weighted SVD proposes an A -orthonormal candidate response span

$$A^{1/2}M_\ell = U_\ell \Sigma_\ell V_\ell^\top, \quad q_i = A^{-1/2}U_{\ell,i}.$$

Keeping the singular directions above a numerical threshold gives a candidate span C_ℓ . We then order directions inside C_ℓ by KRR usefulness. If C_ℓ is represented by an A -orthonormal matrix, set

$$W_\ell = A^{1/2}C_\ell, \quad Z = K_t A^{-1/2}.$$

The eigenvectors of

$$W_\ell^\top Z^\top Z W_\ell$$

ordered from largest to smallest define nested prefixes

$$Q_{\ell,0} \subset Q_{\ell,1} \subset \dots \subset C_\ell.$$

For each prefix we form

$$T_{Q_{\ell,k}} = K_t Q_{\ell,k} Q_{\ell,k}^\top$$

and evaluate

$$\rho_G(T_{Q_{\ell,k}}) = \frac{\|(T - T_{Q_{\ell,k}})A^{1/2}\|_F^2}{R_{\text{KRR}} + \eta_{\text{num}}}.$$

The two ranks in the experiment have different meanings. The prediction-risk rank $r_T(G; 0.05)$ is intrinsic to the finite KRR task: it is computed from $C_G = K_t A^{-1/2}$ before looking at the transformer. It is the smallest optimal rank that leaves at most five percent of KRR posterior risk in the discarded tail. By contrast, the layerwise rank-to-five-percent

$$k_\ell^{5\%} := \min\{k : \rho_G(T_{Q_{\ell,k}}) \leq 0.05\}$$

is representation-dependent: it asks how many directions from the transformer’s candidate response span C_ℓ are needed to reach the same risk tolerance. If $k_\ell^{5\%} \approx r_T$, then the layer contains an almost optimally aligned KRR subspace. If $k_\ell^{5\%} \gg r_T$, the span contains relevant information but in a poorly aligned form. If no such k exists, that layer’s candidate span is not sufficient. We also report $\rho_G(T_{Q_{\ell,r_T}})$, the excess risk of using exactly the intrinsic task rank, and the best error achievable inside C_ℓ .

7 Experiment 1: Final-State Operator Certificate

Experiment 1 is the endpoint certificate. It asks two questions. First, does the trained transformer locally behave like exact KRR at the final prediction output? Second, if it does, is there a compact final-state context-position subspace whose induced reduced ridge operator explains that behavior?

For each episode, the geometry G is fixed and the exact ridge operator

$$T = K_t A^{-1}$$

is computed. The experiment draws separate build probes and held-out evaluation probes. Build probes are used only to extract bases or fit controls; all reported operator errors are evaluated after the basis is frozen. In the reported run there are 4 episodes, 16 build probes, 32 held-out evaluation probes, finite-difference scale $\varepsilon = 10^{-3}$, and a predeclared extraction rank cap $r_{\text{max}} = 9$.

The experiment constructs two final-state bases. The raw basis uses the unperturbed final context residual stream

$$M_{\text{raw}} := H_L^{\text{ctx}} \in \mathbb{R}^{n_{\text{ctx}} \times D}.$$

Each residual coordinate is a context-position pattern, and the A -weighted SVD of M_{raw} gives an A -orthonormal basis Q_{raw} .

The response-only basis uses only the part of the final residual stream that moves under label perturbations. For task-label build probes $u_1, \dots, u_m$, define

$$Z_L(u_j) = \frac{H_L^{\text{ctx}}(y + \varepsilon u_j) - H_L^{\text{ctx}}(y - \varepsilon u_j)}{2\varepsilon}, \quad M_{\text{resp}} = [Z_L(u_1), \dots, Z_L(u_m)].$$

An A -weighted SVD of M_{resp} gives Q_{resp} . This is a stricter availability test than the raw basis: the directions must be visible in label-induced hidden-state changes, not merely present in the final residual stream.

For either frozen basis Q , the KRR-structured reduced operator is

$$T_Q = K_t Q Q^\top.$$

The experiment evaluates both subspace sufficiency,

$$\rho_G(T_Q),$$

and model-subspace alignment,

$$E_\varepsilon(F, T_Q).$$

It also includes a same-span least-squares readout control, denoted T_{actLR} . For any perturbation u , both T_Q and this control first form the same activation coordinates

$$z = Q^\top u.$$

The KRR-structured reduced operator then uses the KRR-implied readout from those coordinates:

$$T_Q u = (K_t Q)(Q^\top u), \quad C_{\text{KRR}} = K_t Q.$$

The control keeps the same features $Q^\top u$, but fits the readout directly to the transformer’s finite-difference responses on build probes $u_1, \dots, u_B$:

$$C_Q \in \arg \min_{C \in \mathbb{R}^{n_{\text{tgt}} \times r}} \sum_{j=1}^B \left\| D_\varepsilon F(y)[u_j] - C Q^\top u_j \right\|_2^2.$$

It then defines

$$T_{\text{actLR}} u = C_Q (Q^\top u), \quad T_{\text{actLR}} := C_Q Q^\top.$$

Thus the two operators use the same Q -features but different readouts: T_Q uses the fixed KRR readout $K_t Q$, while T_{actLR} uses the least-squares readout C_Q fitted to the transformer. This control is important because it separates two questions. A good T_{actLR} fit would show that the transformer’s local response is decodable from the activation coordinates. A good T_Q fit shows something stronger: the same coordinates support the KRR-constrained readout.

The behavioral check succeeds on task-label probes. The pointwise prediction is already close to KRR, with relative model–KRR discrepancy 0.01045, and the held-out finite-difference operator error is

$$E_\varepsilon(F, T) = 0.01207.$$

The additivity defect is much smaller than this error: 3.96×10^{-4} on task probes and 3.91×10^{-3} on isotropic probes at $\varepsilon = 10^{-3}$, with comparable values at 10ε . Thus the local response is sufficiently close to additive for the operator comparison to be meaningful.

The final-state subspace certificates also succeed on task probes. For the raw basis,

$$\rho_G(T_Q) = 0.00279, \quad E_\varepsilon(F, T_Q) = 0.01194.$$

For the response-only basis,

$$\rho_G(T_Q) = 0.01750, \quad E_\varepsilon(F, T_Q) = 0.01239.$$

So the final residual stream exposes compact context-position subspaces whose induced reduced ridge operators add less than five percent of KRR posterior risk, and the transformer’s local response follows those reduced operators.

The T_{actLR} control shows what this certificate is not saying. On task probes, T_{actLR} fits the transformer slightly better than T_Q : for the raw basis

$$E_\varepsilon(F, T_{\text{actLR}}) = 0.00729 \quad \text{versus} \quad E_\varepsilon(F, T_Q) = 0.01195.$$

But it agrees worse with exact KRR:

$$\rho_G(T_{\text{actLR}}) = 0.16040 \quad \text{versus} \quad \rho_G(T_Q) = 0.00279.$$

The response-only basis shows the same pattern. Therefore the certificate is not merely that a free low-rank readout can fit the model response; it is that the same activation-derived subspace supports the KRR-structured operator. Matched random subspaces of the same rank fail the operator-sufficiency test by large factors, with $\rho_G(T_Q) \approx 15.9$ for the raw rank and $\rho_G(T_Q) \approx 11.7$ for the response rank.

The limitation appears immediately off distribution. On isotropic probes,

$$E_\varepsilon(F, T) \approx 0.302.$$

Thus the model is no longer KRR-like on arbitrary label directions. Experiment 1 therefore supports a task-local final-state operator certificate, not a global implementation of ridge regression on all context-label directions. The subspace sufficiency claim remains stated in the task-distributed prediction-risk metric ρ_G .

Table 1: Experiment 1 predictive and behavioral checks, mean over 4 episodes.

Quantity	Transformer	Exact KRR	Transformer-KRR
Pointwise MSE to noiseless teacher targets	0.0161	0.0148	–
Pointwise relative discrepancy	–	–	0.01045
Held-out task operator error $E_\varepsilon(F, T)$	–	–	0.01207

Table 2: Experiment 1 additivity defect, mean over 4 episodes. The defect measures the normalized failure of finite-difference responses to add: $(\sum_j \|D_\varepsilon F[u_j + v_j] - D_\varepsilon F[u_j] - D_\varepsilon F[v_j]\|^2 / (\sum_j \|T(u_j + v_j)\|^2 + \eta_{\text{num}}))^{1/2}$.

Probe distribution	$\varepsilon = 10^{-3}$	$\varepsilon = 10^{-2}$
Task-label probes	3.96×10^{-4}	3.43×10^{-4}
Isotropic probes	3.91×10^{-3}	2.84×10^{-3}

The original plotting diagnostics are omitted here because those panels used the superseded finite-probe response-energy normalization for subspace sufficiency. The retained table reports the same certificate with the prediction-risk excess ρ_G , which is the metric used throughout the revised paper.

Table 3: Reduced operator vs. same-span response-fitted low-rank control, mean over 4 episodes. $E_\varepsilon(F, S)$ denotes transformer-to- S finite-difference operator error on task probes; $\rho_G(S)$ is exact prediction-risk excess relative to KRR.

Basis	$E(F, T_Q)$	$E(F, T_{\text{actLR}})$	$\rho_G(T_Q)$	$\rho_G(T_{\text{actLR}})$
raw	0.01195	0.00729	0.00279	0.16040
response	0.01239	0.01173	0.01750	0.31705

8 Experiment 2: Prediction-Risk Rank and Layerwise Availability

The second experiment asks how prediction-relevant KRR directions appear across layers. In the revised diagnostic, each layer ℓ produces a candidate response span C_ℓ from $M_\ell = D_y H_\ell^{\text{ctx}}$. Inside that span, directions are ordered by their contribution to the KRR operator, and prefixes $Q_{\ell,k}$ are evaluated by the risk-normalized reduced-operator loss

$$\rho_G(T_{Q_{\ell,k}}) = \frac{\|(T - T_{Q_{\ell,k}})A^{1/2}\|_F^2}{R_{\text{KRR}} + \eta_{\text{num}}}.$$

The rank r_T is therefore a target dimensional budget, not a denominator for a dimension ratio. The main question is:

$$\rho_G(T_{Q_{\ell,r_T}}) \leq 0.05?$$

We first rerun this new Experiment 2 on the small linear d_x checkpoints from the original target-rank sweep. The geometry is $n_{\text{ctx}} = 47$, $n_{\text{tgt}} = 16$, with 16 episodes per checkpoint. For these easy linear tasks, the prediction-risk rank equals the input dimension up to the target cap. The layerwise sufficiency diagnostic finds the KRR-sufficient source subspace essentially immediately.

Table 4: Revised Experiment 2 on small linear checkpoints. The response basis is built from $D_y H_\ell^{\text{ctx}}$. “First layer” is the first layer whose response span reaches $\rho_G(T_Q) \leq 0.05$. “Final rank to 5%” is $k_L^{5\%}$, not the intrinsic task rank r_T , although the two coincide in these simple linear runs.

Checkpoint	r_T	first layer	final cand. dim.	final rank to 5%	$E(F, T)$
$d_x = 3$	3.0	0	27.5	3.0	0.00617
$d_x = 5$	5.0	0	27.6	5.0	0.00918
$d_x = 8$	8.0	0	40.1	8.0	0.01250
$d_x = 10$	10.0	0	40.4	10.0	0.01433
$d_x = 15$	14.9	2	46.6	14.9	0.01735

This table should not be read as saying that all computation is already complete at layer 0. It says something narrower: in simple linear tasks, the context-side subspace needed to build the KRR operator is already linearly available in the early label/input response geometry. This is plausible: the labels are injected directly into context positions, and the KRR-relevant source directions are low-dimensional functions of the fixed linear geometry. Later layers still matter for the actual model map F , but the containment question saturates early.

This is why the nonlinear RBF checkpoint is the more informative emergence test. In the RBF $n_{\text{ctx}} = 128$, $n_{\text{tgt}} = 128$ checkpoint, $r_T \approx 24.75$, and the response span does not start sufficient. The main transition occurs between layers 1 and 2:

Thus, for the nonlinear task, the KRR-relevant response subspace mostly emerges by layer 2, becomes essentially sufficient by layers 3–4, and then stabilizes. The final response span has dimension about 119, but the selected KRR-sufficient prefix has dimension about 25, matching

Table 5: Revised Experiment 2 on the RBF 128×128 checkpoint, response basis. The column “rank to 5%” is $k_\ell^{5\%}$, the representation-dependent prefix rank required to reach $\rho_G \leq 0.05$.

Layer	cand. dim.	rank to 5%	$\rho_G(T_{Q_\ell, r_T})$	best excess
0	16	–	1.147	1.147
1	51	38.0	0.100	0.074
2	84	26.9	0.058	0.016
3	101	25.5	0.052	0.006
4	107	25.2	0.050	0.004
5	113	25.1	0.049	0.002
6	103	25.4	0.050	0.004
7	126	24.9	0.048	0.000
8	119	24.9	0.048	0.001

r_T . The hidden response is not compressed to the task rank; rather, a task-rank KRR subspace is present inside a larger response span.

This is the final role of Experiment 2: it is an availability diagnostic, not an individual intervention diagnostic. It asks whether a KRR-sufficient subspace is present in the layerwise response span and at what layer it becomes available. Causal use is tested separately in Experiment 3.

8.1 Dimension-40 Regime-Mismatch Stress Test

The $d_x = 40$ checkpoint separates availability from use. It was trained in a target-poor regime:

$$n_{\text{tgt}} = 8, \quad \tau(\Sigma) = \sum_{i=1}^{40} \frac{\lambda_i}{\lambda_i + \sigma^2} \leq 16,$$

with n_{ctx} sampled as a multiple of the effective dimension. The diagnostic instead uses

$$n_{\text{ctx}} = 47, \quad n_{\text{tgt}} = 40,$$

so it asks for a richer simultaneous target operator than the checkpoint saw directly in one training episode. The checkpoint README reports

$$\text{MSE}/\text{MSE}_{\text{KRR}} = 1.638,$$

so this is not an exact-KRR solver; the question is whether the remaining error is explained by missing KRR source directions or by imperfect use of directions that are already present.

Table 6: Final Experiment 2 on the $d_x = 40$ checkpoint with the native sampler and target-rich diagnostic $n_{\text{ctx}} = 47, n_{\text{tgt}} = 40$. The final-layer rows compare response-only, raw, and combined candidate spans.

Basis	cand. dim.	r_T	$k_L^{5\%}$	reach	$\rho_G(T_{Q_L, r_T})$	best ρ_G	$E(F, T)$
response	40.0	21.0	21.3	0.50	0.118	0.086	0.0939
raw	39.3	21.0	21.6	1.00	0.047	0.007	0.0939
combined	40.0	21.0	21.5	0.75	0.077	0.043	0.0939

The raw final residual stream contains a KRR-sufficient source subspace at the prediction-risk rank: $\rho_G(T_{Q_L, r_T}) \approx 0.047$, just below the five-percent tolerance. The response-only span is close but not reliably sufficient at the same rank, with $\rho_G \approx 0.118$ and reach 0.50. Thus the target-rich failure is not simply absence of the relevant directions. A KRR-oriented subspace is visible in the

residual stream, but the label-induced response geometry and the realized model map do not stably use it as the exact target-rich KRR operator.

The model-side error confirms this reading:

$$E(F, T) \approx 0.0939, \quad \frac{\mathbb{E}\|(F - T)y\|^2}{R_{\text{KRR}}} \approx 0.132.$$

These values are too large for a converged KRR mechanism and are qualitatively consistent with the README-level MSE ratio 1.638. Mechanistically, target-poor training can still optimize population prediction in expectation, but it gives weaker per-episode pressure to expose and route a stable target-rich label-response operator. The $d_x = 40$ run is therefore a regime-mismatch stress test: the right source subspace is largely present, especially in the raw final state, while the learned readout/routing remains imperfect.

This motivates the sequence-length prescription above. If the goal is to train and certify a true prediction-relevant KRR mechanism in a given regime, choose n_{ctx} and n_{tgt} so that

$$\min\{n_{\text{ctx}}, n_{\text{tgt}}, r_*(n_{\text{ctx}}; 0.05)\} = r_*(n_{\text{ctx}}; 0.05)$$

during training as well as diagnosis. Target-rich diagnostics alone can reveal the mismatch; target-rich training is the direct way to put pressure on the model to learn the full visible map.

8.2 RBF Kernel Experiment: Prediction-Risk Rank and Availability

The RBF experiment tests whether the same operator picture survives when the prediction-risk rank is no longer tied to the input dimension. The checkpoint is trained on labels from an RBF Gaussian process with lengthscale 3.0, signal variance 1.0, and noise variance $\sigma^2 = 0.1$. On held-out RBF tasks the model is close to KRR: at the end of training the prediction MSE is within about 1–3% of exact KRR, and the model–KRR prediction discrepancy is around 10^{-3} . The important change is that the rank is now set by prediction risk:

$$r_T = \min \left\{ r : \sum_{i>r} s_i^2 \leq 0.05 R_{\text{KRR}} \right\}.$$

This number is computed from the KRR geometry before looking at activations. It asks how many modes are needed so that discarding the rest would add at most five percent of the KRR posterior prediction risk. For the clean 128×128 RBF checkpoint, this gives

$$r_T \approx 24.75.$$

Table 7: RBF 128×128 diagnostics from the regenerated final suite. The layerwise response basis is the Experiment 2 availability basis.

Diagnostic	Dimensional scale	Error scale	Reading
Model response	$r_T = 24.75$	$E(F, T) = 0.0306$	near-KRR behavior
Model excess risk	$r_T = 24.75$	$\rho(F) = 0.0203$	below 5% risk tolerance
Final response span	selected rank 24.9	$\rho(T_Q) = 0.048$	sufficient at r_T
Final raw span	selected rank 26.0	$\rho(T_Q) = 0.054$	near sufficient

The table gives the clean nonlinear conclusion. The model itself is close to KRR, and the final response span contains a rank-25-scale subspace whose induced reduced KRR operator reaches the predeclared five-percent prediction-risk tolerance. This is why taking the final hidden representation

and allowing the selected rank to match r_T recovers the operation: the directions are present, but they are distributed inside a larger response span and must be oriented by KRR relevance.

We also reran the revised layerwise diagnostic on the older fixed- $\ell = 3$ RBF checkpoint `rbf_fixed_13_original.pt`, at $n_{\text{ctx}} = 47$. This checkpoint was originally diagnosed in a target-restricted setting, so we evaluate the same model at $n_{\text{tgt}} = 16, 64, 128$. This separates the intrinsic task rank r_T from the representation-dependent final response rank $k_L^{5\%}$.

Table 8: Revised Experiment 2 on the older $n_{\text{ctx}} = 47$ fixed- $\ell = 3$ RBF checkpoint. Here r_T is the intrinsic prediction-risk rank of the finite KRR task, while $k_L^{5\%}$ is the number of final response-span directions needed to reach $\rho_G \leq 0.05$.

n_{tgt}	r_T	$k_L^{5\%}$	$\rho_G(T_{Q_L, r_T})$	best ρ_G	$E(F, T)$
16	10.6	11.1	0.049	0.009	0.067
64	17.1	17.8	0.053	0.015	0.074
128	18.3	19.3	0.057	0.021	0.097

The table refines the earlier target-restriction story. The older checkpoint does not simply lack the larger KRR-relevant directions: $k_L^{5\%}$ tracks r_T closely even as the target set grows. Thus the final response representation contains a near-sufficient KRR subspace. What degrades is the realized model response: $E(F, T)$ grows as the diagnostic exposes a richer target-visible operator. The failure mode is therefore readout/routing alignment, not absence of a KRR subspace. The 128×128 checkpoint remains the cleaner nonlinear result because training and diagnosis are matched in the richer regime and behavior, final-state availability, and prediction-risk rank are aligned in the same geometry.

The primary object in Experiment 2 is therefore the layerwise response span, ordered by KRR-targeted energy and evaluated by prediction-risk excess. Under that definition:

RBF behavior and final-state availability succeed,

and the older $n_{\text{ctx}} = 47$ checkpoint shows a softer failure mode: the response representation remains near-sufficient, but the realized model response degrades as the target set is made richer.

9 Experiment 3: Causal Direction Removal

The third experiment asks whether the task-important directions inside Q are actually used by the transformer. This is necessary because decodability and sufficiency do not by themselves prove causal use.

Start with a raw final basis $Q = [q_1, \dots, q_r]$. For each direction, form Q_{-j} by removing that column. The point of the leverage score is to rank directions by their importance to the KRR-constrained reduced operator, not by activation magnitude. A direction should count as task-important if deleting it changes T_Q 's response on task-label probes. Define

$$\Lambda_j^{\text{red}}(U) := \left(\frac{\sum_{u \in U} \|(T_Q - T_{Q_{-j}})u\|_2^2}{\sum_{u \in U} \|Tu\|_2^2 + \eta_{\text{num}}} \right)^{1/2}.$$

A direction has high task leverage if this score is large on task-label probes. This ranking is what connects the causal intervention to the operator certificate: the surgery removes directions that T_Q itself says are important for task-distributed ridge responses.

The intervention removes selected directions from the context-token residual stream:

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - \sum_{j \in J} q_j q_j^\top A H_s^{\text{ctx}},$$

then rolls the remaining layers forward. Matched controls remove low-task-leverage directions from the same basis, high-isotropic-but-low-task directions, high-activation-variance-but-low-task directions, and random subsets.

Before surgery, the checkpoint is in the task-local certificate regime:

$$E_{\text{task}}(F, T) = 0.00926, \quad E_{\text{task}}(F, T_Q) = 0.00920, \quad \rho_G(T_Q) = 0.00149.$$

The same episodes retain the off-distribution drift seen in Experiment 1:

$$E_{\text{iso}}(F, T) = 0.266.$$

At residual state $s = 1$, removing a single high-task-leverage direction causes prediction drift 4.68, KRR-error inflation 649 $\times$, and task-response damage 41.59. Removing a single low-task-leverage direction from the same basis is nearly inert: prediction drift $9.8 \cdot 10^{-4}$, KRR-error inflation 1.03 $\times$, and task-response damage 0.0024. The layer sweep localizes the effect: high-task surgery is most damaging at early residual states, before later attention has routed the relevant context information into target tokens, and context-only surgery after the final layer is a no-op. At $s = 1$, aggregate task leverage and task response damage are strongly related across matched non-random controls (log-log Pearson 0.91, Spearman 0.87).

The complement ablation sharpens the mediation claim. At the same residual state $s = 1$, keeping only the A -projection of context residual states onto Q ,

$$H_s^{\text{ctx}} \mapsto Q Q^\top A H_s^{\text{ctx}},$$

nearly preserves the task-local response over 8 episodes:

$$E_{\text{task}}(F, T) = 0.01836, \quad \text{response damage} = 0.01634.$$

Removing Q instead,

$$H_s^{\text{ctx}} \mapsto H_s^{\text{ctx}} - Q Q^\top A H_s^{\text{ctx}},$$

destroys the response:

$$E_{\text{task}}(F, T) = 106.46, \quad E_{\text{task}}(F, T_Q) = 106.46, \quad \text{response damage} = 106.46.$$

So the complement of Q is largely dispensable for the task-local KRR response in this intervention family, while Q itself is necessary.

Whole-subspace intervention at residual state $s = 1$

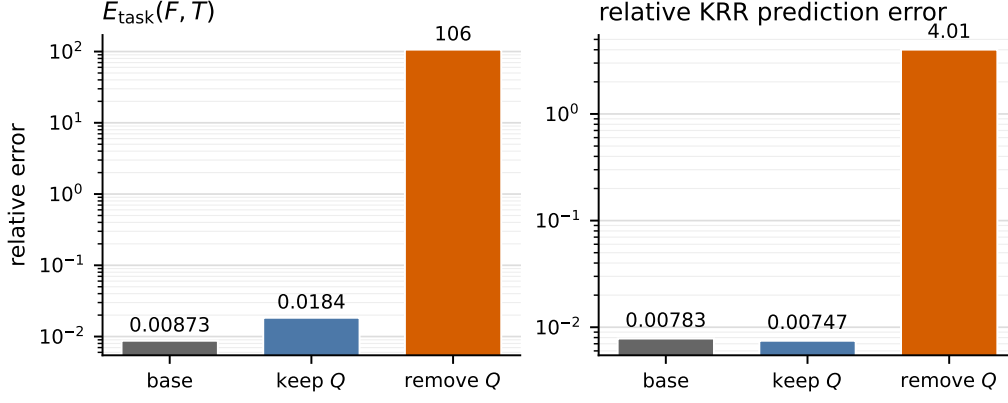


Figure 1: Experiment 3: whole-subspace complement ablation at residual state $s = 1$. Left panel: task finite-difference response error $E_{\text{task}}(F, T)$. Right panel: relative KRR prediction error $\|F_{\text{surg}}(y) - Ty\|_2 / \|Ty\|_2$. Keeping only the Q -projection nearly preserves both quantities; removing Q destroys both.

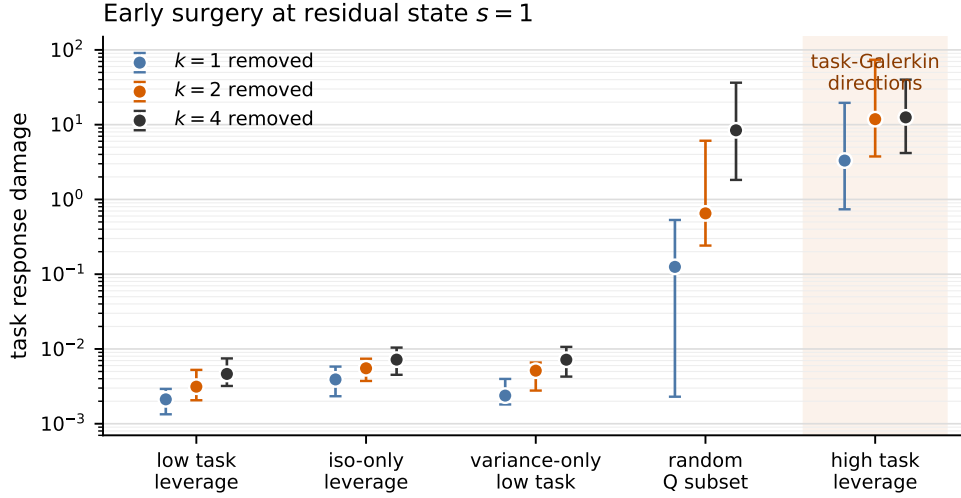


Figure 2: Experiment 3: causal effect of task-important activation directions. The plotted metric is task finite-difference response damage $(\sum_j \|D_\varepsilon F_{\text{surg}}(y)[u_j] - D_\varepsilon F_{\text{base}}(y)[u_j]\|^2 / (\sum_j \|Tu_j\|^2 + \eta_{\text{num}}))^{1/2}$, for task probes $u_j \sim \mathcal{N}(0, A)$. High task-leverage directions produce orders-of-magnitude larger damage than low-task-leverage directions from the same activation basis.

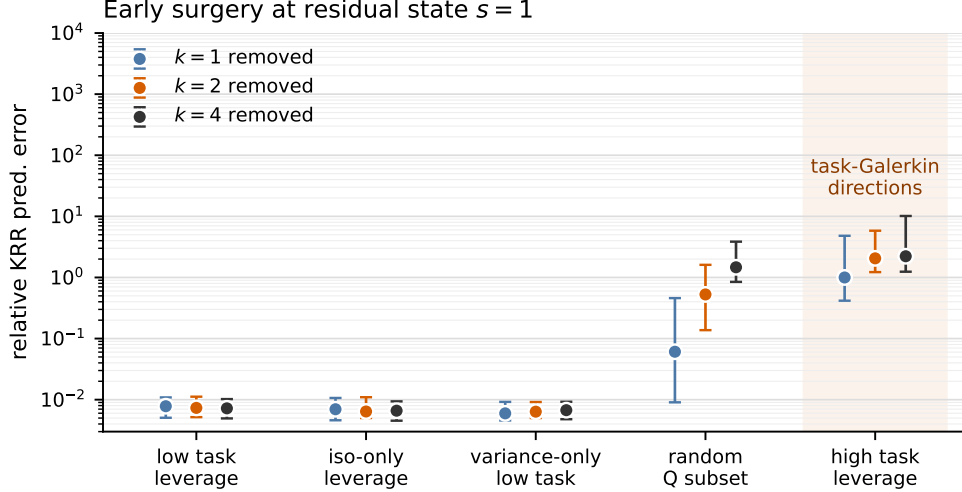


Figure 3: Experiment 3: the same surgery plot, but measuring relative KRR prediction error $\|F_{\text{surg}}(y) - Ty\|_2 / \|Ty\|_2$ instead of finite-difference response damage. High task-leverage removals also destroy the model’s prediction, while low-task-leverage removals are nearly inert.

This supports the causal-use claim in an operational sense. The directions that the reduced KRR operator identifies as task-important are not merely decodable; removing them selectively damages the transformer’s task-local response, while matched controls do not.

10 What Is Justified

The experiments justify a bounded mechanistic claim:

For the tested checkpoint, geometries, perturbation scale, and task-label probe distribution, the transformer’s local label response is well approximated by a reduced KRR operator induced by a compact context-side activation subspace. In successful settings, layerwise response spans contain a prediction-risk-rank subspace whose induced reduced operator is close to exact KRR; in the original linear setting, high-task-leverage directions from this subspace are causally important for the transformer’s response.

They do not justify stronger claims. They do not show that the transformer globally equals ridge regression. They do not show that derivative matching at one sampled label vector implies equality on a neighborhood. They do not show that the transformer literally executes the explicit algorithm

$$y \mapsto QQ^\top y \mapsto K_t QQ^\top y$$

as its internal program. They also do not show that architecture alone guarantees the behavior. The prediction-risk rank calculation supplies a requirement for context-side subspaces; it is not an SGD theorem.

The clean final formulation is:

The trained transformer locally realizes the ridge-regression label-response operator on task-distributed directions. In the tested regime, this behavior is explained by a compact context-side activation subspace whose induced reduced ridge operator approximates

exact KRR at the prediction-risk rank. The layerwise diagnostic tracks when such a subspace becomes available; the causal intervention then asks whether the model actually uses the selected directions.

11 Logical Chain

The argument can be read as a sequence of commitments, each narrower than the previous one:

1. Freeze the episode geometry $G = (X_{\text{ctx}}, X_{\text{tgt}})$.
2. Under fixed G , ridge regression becomes $T = K_t A^{-1}$.
3. Treat the transformer as a label map $F : \mathbb{R}^{n_{\text{ctx}}} \rightarrow \mathbb{R}^{n_{\text{tgt}}}$.
4. Estimate the local response $D_\varepsilon F(y)[u]$.
5. Test behavior by checking $E_\varepsilon(F, T)$ on task-label probes.
6. Compute the prediction-risk rank r_T from $C_G = K_t A^{-1/2}$ and R_{KRR} .
7. Extract activation-derived context-side candidate spans C_ℓ .
8. Order each candidate span by KRR-targeted energy and form prefixes $Q_{\ell,k}$.
9. Construct the KRR-constrained reduced operator $T_Q = K_t Q Q^\top$.
10. Test subspace sufficiency with $\rho_G(T_Q)$.
11. Test model-subspace alignment with $E_\varepsilon(F, T_Q)$.
12. In causal runs, remove high-leverage Q -directions and check whether task-local response is damaged.

This is the full mechanism story: behavioral similarity, a KRR-structured activation certificate, prediction-risk-rank sufficiency, layerwise emergence of the relevant subspace, and causal specificity of high-leverage directions where the intervention has been validated.